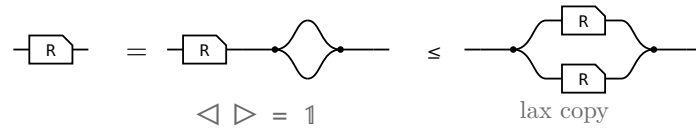


The allegory axioms: the proofs

The companion to `allegory-axioms.typ`, which states these laws and their pictures. Here each one is worked: the steps of the Lean proof, side by side, with the rule that reaches each under it.

§1. $R \circ R = R$, the one that is not bookkeeping



The last picture is $R \circ R$ itself, so this is $R \leq R \circ R$ and the lax copy law is the whole of it. The other direction is not proved here: $R \circ S \leq R$ holds for every S , by weakening S to $\top$ and then collapsing $R \circ \top$ — which is the paper’s own remaining three steps, packaged once.

§2. The modular law, from Frobenius

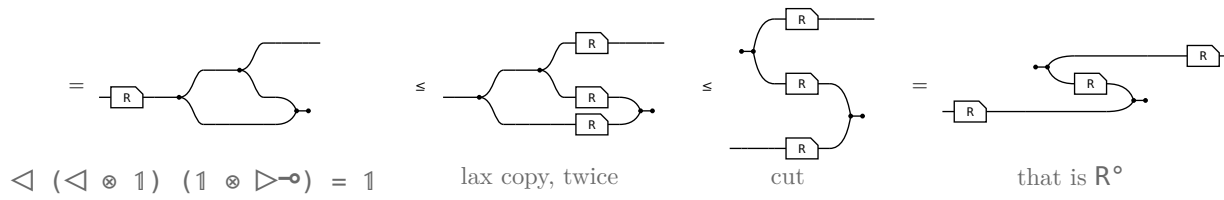
the modular law, $R \circ S \circ T \leq (R \circ T \circ S^\circ) \circ S$		
term	picture	the rule that reaches it
$(R \circ S) \circ T$		the start: $R \circ S \circ T$.
$= (\triangleleft (R \otimes T)) ((S \otimes 1) \triangleright)$		Factor out the shared prefix: $(R \circ S) \otimes T$ is $(R \otimes T) (S \otimes 1)$.
$\leq (\triangleleft (R \otimes T)) ((1 \otimes S^\circ) (\triangleright S))$		The merge slide. S slides through the merge and reappears as S° on the other strand. The only inequality in the proof.
$= (R \circ (T \circ S^\circ)) \circ S$		Fold the prefix back in; the copy-merge pair reads as a meet again.

S occurs once on the left of the slide and twice on the right, and the lax copy law is the only law in the definition that may duplicate a box — so that is where the inequality has to be.

the merge slide, $(S \otimes 1) \triangleright \leq (1 \otimes S^\circ) \triangleright S$		
term	picture	the rule that reaches it
$(S \otimes 1) \triangleright$		the start: $(S \otimes 1) \triangleright$.

§4. The same, from the cap end

The section above cuts twice in its chain. This one cuts once and pays for the other cut with the **lax copy law** — $R \triangleleft \leq \triangleleft (R \otimes R)$, the only law in the definition that may duplicate a box. Read 1 as a copy tree whose last two legs are capped back:

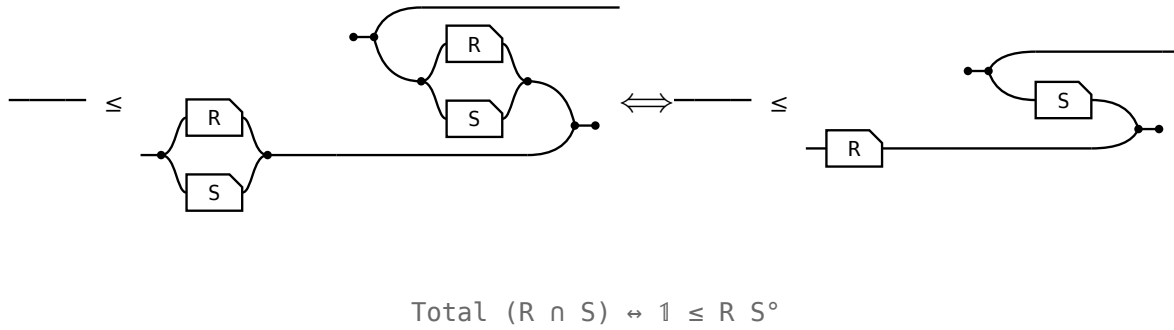


The three boxes are not there to begin with: there is one, and the two copy dots walk back **through** it, leaving it behind three times. So the $\triangleright \circ$ is the frame's own cap before anything is cut, and the copy tree is the only thing left to cut.

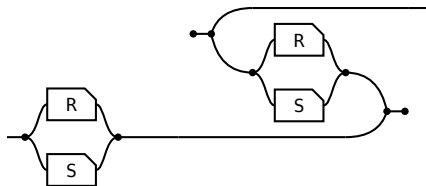
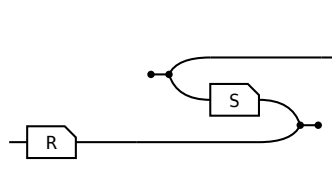
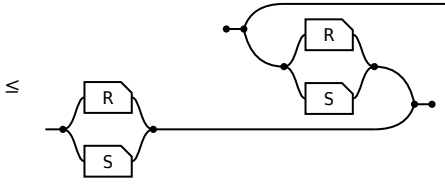
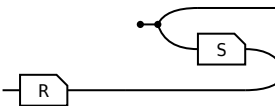
Neither chain is cheaper in what it **spends**. $\triangleleft \triangleright = 1$ is itself the merge cut, so the second $1 \leq \circ \circ$ is underneath this one; $R \cap R = R$ is the lax copy law, so that is underneath the section above. Both routes pay both. What differs is the **layer**: this chain names only the copy, the cap and the box, so it holds as soon as the converse does — before $\cap$ exists. The section above climbs into the meet semilattice and back down to state a containment in which no meet occurs.

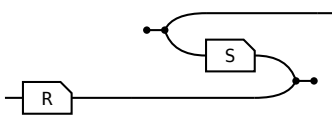
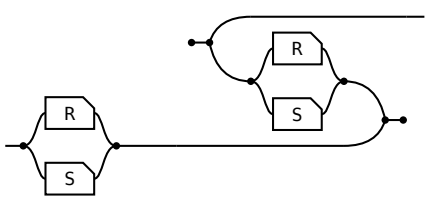
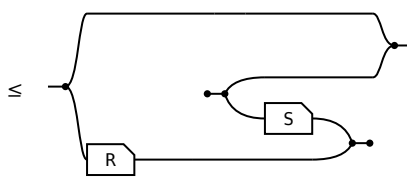
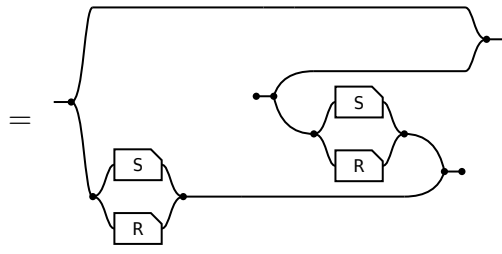
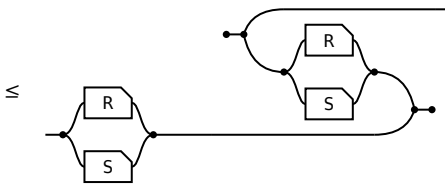
§5. Entireness of an intersection

R is **entire** when $1 \sqsubseteq R R^\circ$. When is an **intersection** entire?



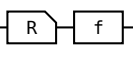
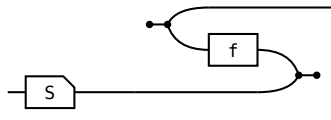
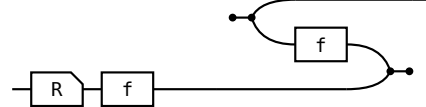
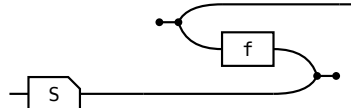
On the left $R \cap S$ is named **twice**; on the right R and S once each and the meet is gone. The meet is needed to **ask** whether something is entire, not to **answer** it. Where $\sqsubseteq$ is **defined** as $X \cap Y = X$ the two sides are one proposition; here $\leq$ is primitive, so both directions are real steps — and they cost very different things.

$\Rightarrow$ given $R \cap S$ entire,  show 		
term	picture	the rule that reaches it
1		the start: 1 .
$\leq (R \cap S) (R \cap S)^\circ$		the hypothesis: $1 \leq P P^\circ$ at $P = R \cap S$.
$\leq R S^\circ$		monotonicity, twice. No modular law.

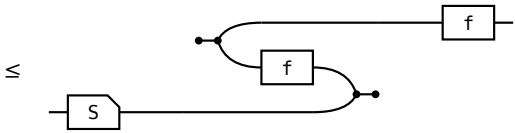
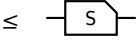
$\Leftarrow$ given $\text{---} \leq$  show $R \cap S$ entire, $\text{---} \leq$ 		
term	picture	the rule that reaches it
1	---	the start: 1.
$\leq 1 \cap (R S^\circ)$	$\leq$ 	the hypothesis, met with 1.
$= 1 \cap ((S \cap R) (S \cap R)^\circ)$	$=$ 	$1 \cap R S^\circ = 1 \cap (S \cap R) (S \cap R)^\circ$. $R S^\circ$ relates a to a' when some b has $a R b$ and $a' S b$; the $1 \cap$ sets $a' = a$, and it reads “some b has $a R b$ and $a S b$ ” — one arrow of $R \cap S$. Two arrows become one, and that is what needs the modular law.
$\leq (R \cap S) (R \cap S)^\circ$	$\leq$ 	$X \cap Y \leq Y$, then $S \cap R = R \cap S$.

§6. The shunting rule

A map moves from one side of a containment to the other at the cost of its converse, $R f \leq S \iff R \leq S f^\circ$. It is how a function gets out of the way so the relation underneath can be reasoned about. Each direction spends exactly one half of **map** — never both.

shunting right, $\Rightarrow$ given $\text{---} \text{---} \leq$  show $\text{---} \leq$ 		
term	picture	the rule that reaches it
R		the start: R.
$\leq R (f f^\circ)$	$\leq$ 	total: $1 \leq f f^\circ$, so $f f^\circ$ may be inserted after R.
$\leq S f^\circ$	$\leq$ 	Re-bracket to $(R f) f^\circ$ and apply the hypothesis.

shunting right, $\Leftarrow$ given  $\leq$   show   $\leq$ 

term	picture	the rule that reaches it
$R \ f$	 	the start: $R \ f$.
$\leq S \ (f^\circ \ f)$		Apply the hypothesis on the left; the two f s are now adjacent.
$\leq S$		single valued: $f^\circ \ f \leq 1$, so the pair cancels.

Shunting left is the mirror, same shape, map on the other side throughout. What the rule is, is an adjunction: $f \dashv f^\circ$, with **total** the unit and **single valued** the counit, so “ f is a map” and “ f has a right adjoint, namely f° ” are one statement.